

# INDIAN STATISTICAL INSTITUTE

Practice Problems, 06 August 2026  
M.TECH(CS/CrS) I YEAR

## INTRODUCTION TO PROGRAMMING

1. We say that a string  $s$  is a “good password” if it is a string of printable ASCII characters with the following characteristics: (i) The string is at least 8 characters long (ii) The string contains both upper and lower case alphabets (iii) The string contains at least one numeral (iv) The string contains at least one character which is neither an alphabet nor a numeral. Given a string, find whether it is a good password.
2. Write a program to count the number of occurrences of each letter A-Z or a-z in the input. The input may span multiple lines. While counting letters, do not distinguish between uppercase and lowercase letters. Ignore all non-letter characters, e.g., digits, punctuation, spaces and tabs. You should print, in alphabetical order, only the letters that actually occur in the input, along with their counts ( $\geq 1$ ).

Sample input:

C Programming is not fun!!

Sample output:

|   |   |
|---|---|
| a | 1 |
| c | 1 |
| f | 1 |
| g | 2 |
| i | 2 |
| m | 2 |
| n | 3 |
| o | 2 |
| r | 2 |
| s | 1 |
| t | 1 |
| u | 1 |
| p | 1 |

3. Run-length encoding (RLE) is a form of lossless data compression in which runs of data (consecutive occurrences of the same data value) are stored as a single occurrence of that data value and a count of its consecutive occurrences, rather than as the original run. For example, the run-length encoded form of `aaabccd` is `a3bc2d`.
  - (a) Write a program to convert a given string  $s$ , consisting of at most 100 letters (lower / upper case), to  $s'$ , its run-length encoded form. This means that  $s'$  will contain the same sequence of distinct characters as  $s$ , but any  $m > 1$  consecutive occurrences of a character will be replaced by a single occurrence of the character immediately followed by the integer  $m$  (in base 10).
  - (b) Write a program to decode a given run-length encoded string  $s'$  to its original form  $s$ . For example, given `Aa3bC2d`, your program should print `AaaabCCd`. Similarly, given `abcd`, your program should print `abcd` itself; given `z10`, your program should print `zzzzzzzzzz`.
4. The International Standard Book Number (ISBN) is a 10-digit code of the form  $d_{10}d_9d_8d_7\dots d_2d_1$  that uniquely specifies a book. The rightmost digit  $d_1$  is a checksum digit that must be chosen to satisfy the following condition:

$$d_1 + 2d_2 + 3d_3 + \dots + 10d_{10} \equiv 0 \pmod{11}$$

Note that the checksum digit  $d_1$  can range from 0 to 10 (inclusive); if  $d_1 = 10$ , it is represented as  $X$ . Write a C program that reads a 9-digit integer, computes the checksum digit, and prints the full 10-digit ISBN.

Sample input: 020131452

Sample output: 0201314525

Explanation: The correct checksum digit is 5, since:

$$5 + 22 + 35 + 44 + 51 + 63 + 71 + 80 + 92 + 100 = 88 \equiv 0 \pmod{11}$$